EXP.NO:14

TWO ELEMENTS SUM TO X

PROBLEM STATEMENT:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

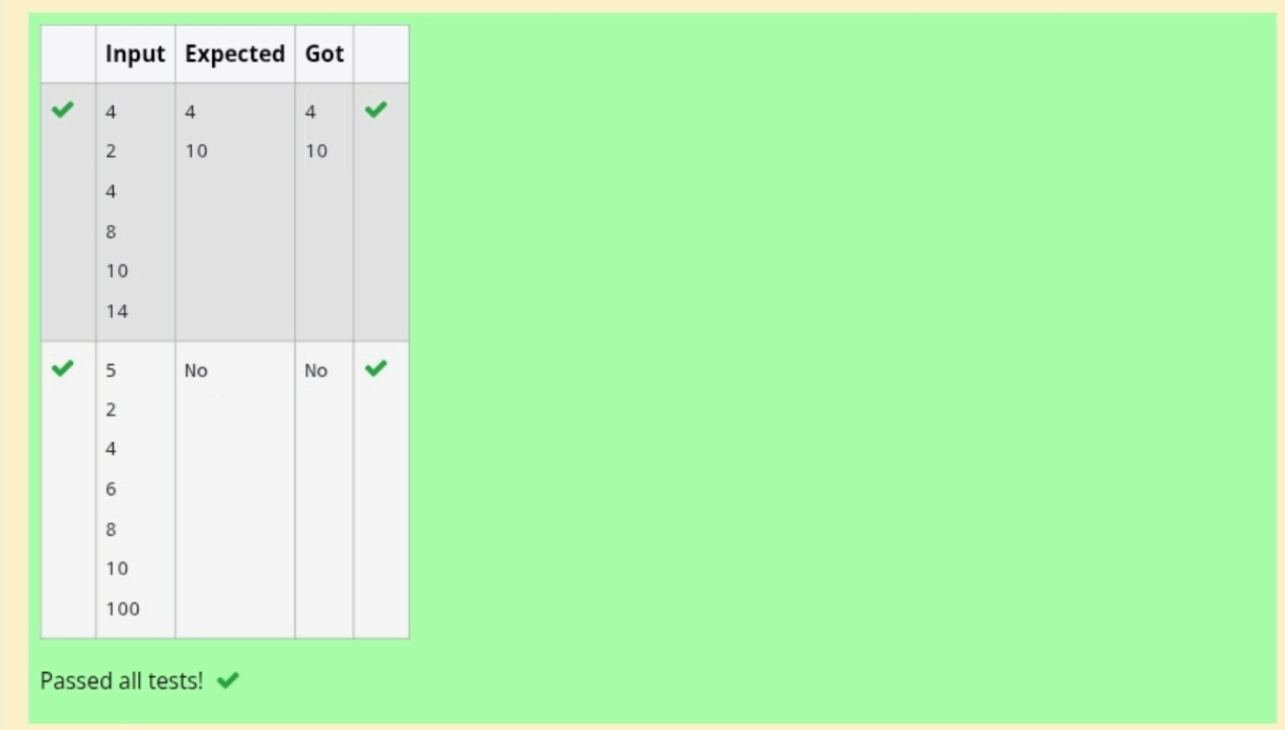
Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

CODE:

```c
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    int sum;
    scanf("%d",&sum);
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(i!=j && a[i]+a[j]==sum){
                printf("%d\n%d",a[i],a[j]);
                return 0;
            }
        }
    }
    printf("No");
}
```

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OUTPUT:



	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

RESULT:

The code is executed successfully.